



Birzeit University

Faculty of Engineering and Technology Department of

Electrical and Computer Engineering

APPLIED CRYPTOGRAPHY

ENCS4320

Homework # 1

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Section: 1

Question #1:

Question 1

To Find the plaintext and key we need to try all possible keys :

K	M
0	W L I M W X L I W S Y P S J Q C W S Y P
1	V K H L V W K H V R X O R I P B V R X O
2	U J G K U V J G U O W N O H O A U O W N
3	T I F J T U I F T P V M N G N Z T P V M
4	S H E I S T H E S O U L O F M Y S O U L

The key is 4 (ROT-4) and the plaintext is "she is the soul of my soul".

Question #2:

Question 2

a) In substitution cipher each plaintext character gets mapped to a fixed ciphertext character so if the output of the encryption has two similar characters then the password is 'bddf' since it has two (dd) in the middle. otherwise, the password is 'wxyz'.

b) In Vigenere Cipher with period 2 the only thing that we know that the first and Third character have the same shift amount while the second and fourth have also the same shift amount.

Assume the cipher text is $g\ Q\ F\ S\ H$
 $16\ 5\ 18\ 7$

b	d	d	f	}	w	x	y	z
1	3	3	5		22	23	24	25
after shift:					after shift:			
15	2	15	2		6	18	6	18
<u> </u>					<u> </u>			

Since in both password the shift amount between the first and third, second and fourth is equal then we can not detect which password was used.

C) In Vigenere cipher with period 3 the only thing we know that the first and forth character have the same shift amount, we can use that to see if we can detect the password.

Assume cipher is E K S H
 $4 \ 10 \ 18 \ 7$

b d d f $1 \ 3 \ 3 \ 5$	{	w x y z <u>$22 \ 23 \ 24 \ 25$</u>
After shift 3		After shift 3
$3 \ 7 \ 15 \ 2$ <u> </u>		$-18 \ -13 \ -6 \ -18$ <u> </u>

Since the shift amount is equal in password 'wxyz' then this is the password used to result the cipher texts, then Vigenere with period 3 can detect the user password.

Question #3:

Question 3

a) 4.2×10^9 cycles per sec per core

for 16 cores $\Rightarrow 4.2 \times 10^9 \times 16 = 67,2 \times 10^9 \text{ C/s}$

$$\text{avg time} = \frac{2^{56} \setminus 20}{67,2 \times 10^9} = 53614,28 \text{ sec}$$

$$\begin{aligned} \text{to convert to Year} &= \frac{53614,28}{60 \times 60 \times 24 \times 365} = \frac{1,7 \times 10^{-3}}{2} \\ &= 8,5 \times 10^{-4} \text{ Year} \end{aligned}$$

$$\text{b) Avg time for } \frac{2^{128} \setminus 20}{67,2 \times 10^9} = 2,532 \times 10^{26} \text{ sec}$$

128 bits key

$$\begin{aligned} \text{to convert to Year} &= \frac{2,532 \times 10^{25}}{60 \times 60 \times 24 \times 365} = \frac{8,023 \times 10^{18}}{2} \\ &= 4,0115 \times 10^{18} \text{ Year} \end{aligned}$$

Question #4:

Question 4

$$a) \quad 3 - 11 \pmod{9} = -8 \pmod{9}$$

$$-8 = 9x + r \quad t = -1$$

$$-8 = -9 + r$$

$$r = 1$$

$$3 - 11 \pmod{9} = \boxed{1}$$

$$b) \quad 15 \times 29 \pmod{13}$$

$$= ((15 \pmod{13}) \times (29 \pmod{13})) \pmod{13}$$

$$(2 \times 3) \pmod{13}$$

$$6 \pmod{13} = \boxed{6}$$

$$c) -12 \setminus 35 \pmod{19} = -12 \times 35^{-1} \pmod{19}$$

$$((-12 \pmod{19}) \times (35^{-1} \pmod{19})) \pmod{19}$$

$$=$$

$$\Rightarrow -12 \pmod{19}$$

$$-12 = 19x + r \quad x = -1$$

$$-12 = -19 + r$$

$$r = 7$$

$$\Rightarrow 35^{-1} \pmod{19}$$

$$35 = 19 \times 1 + 16$$

$$19 = 16 \times 1 + 3$$

$$16 = 3 \times 5 + 1$$

now let go back to have an equation
where $1 = r \times 35 + t \times 19$?

$$1 = 16 - 3 \times 5$$

$$1 = 16 - (19 - 1 \times 16) \times 5$$

$$1 = 16 + 16 \times 5 - 19 \times 5$$

$$1 = 16 \times 6 - 19 \times 5$$

$$1 = (35 - 1 \times 19) \times 6 - 19 \times 5$$

$$1 = 6 \times 35 - 11 \times 19$$

$$35^{-1} \bmod 19 \equiv 6 \times 35 \equiv 1 \bmod 19$$

$$(7 \times 6) \bmod 19$$

$$42 \bmod 19 = \boxed{4}$$

d) For them to be co-prime $\gcd \stackrel{??}{=} 1$

$$\gcd(172, 68) = \gcd(68, 172 \bmod 68)$$

$$= \gcd(68, 36) = \gcd(36, 68 \bmod 36)$$

$$= \gcd(36, 32) = \gcd(32, 36 \bmod 32)$$

$$= \gcd(32, 4) = \gcd(4, 32 \bmod 4)$$

$$= \gcd(4, 0) = 4 \neq 1$$

They are not co-prime numbers.

Question #5:

Question 5

need to check if the ciphertext leak any information about the plaintext or not.

K	m	C
0	0	0
1		1
2		2
3	1	0
0		1
1		2
2	2	0
3		1
0		2
1		0
2		1
3		2

to check if is perfectly secret

$$P[M = m | C = c] = P[M = m]$$

$$P[M = 0 | C = 2] \stackrel{??}{=} P[M = 0]$$

$$\frac{1}{4} \neq \frac{1}{3}$$

then it not perfectly secret.

Question #6:

Question 6

$$(EF)_{16} = 1110 \ 1111$$

$$(A4)_{16} = 1010 \ 0100$$

$$(D3)_{16} = 1101 \ 0011$$

$$\textcircled{1} EF \oplus A4 = 0100 \ 1011$$

$x_6 x_5 = 10 \rightarrow$ then p_1 and p_2 is either a lower character or digit

$x_4 = 0 \rightarrow \begin{matrix} \nearrow 1 \oplus 1 \\ \searrow 0 \oplus 0 \end{matrix} \rightarrow$ For the digit since space not allowed, then lower-case letter is in range $['p', 'z']$.

$\begin{matrix} x & x & x & x \\ 3 & 2 & 1 & 0 \end{matrix} = 1011 \rightarrow (B)_{16}$, since the last four bits not equal in p_1 and p_2 .

$$\textcircled{2} \text{EF} \oplus \text{D3} = 0011\ 1100$$

$x_6 x_5 = 01 \rightarrow$ then p_1 and p_3 is either lower-case letter or upper-case letter

$x_4 = 1 \begin{matrix} \rightarrow 0 \oplus 1 \\ \rightarrow 1 \oplus 0 \end{matrix}$, since p_1 is either digit or

lower-case in range ['p', 'z'] then we should

choose so that $1 \rightarrow$ is for the lower-case and

$0 \rightarrow$ is for the upper-case letter in range ['A', 'O'].

$\begin{matrix} x & x & x & x \\ 3 & 2 & 1 & 0 \end{matrix} = 1100 \rightarrow (C)_{16}$, since the last four bits

in both are not equal.

$$\textcircled{3} \quad A_4 \oplus D_3 = 01110111$$

$x_6 x_5 = 11 \rightarrow$ then p_2 and p_3 is either a digit or an upper-case letter.

$x_4 = 1 \begin{cases} 0 \oplus 1 \\ 1 \oplus 0 \end{cases}$, since p_2 is either digit, and p_3 is a upper-case letter, then $0 \rightarrow$ for the digit and $1 \rightarrow$ for the upper-case then p_3 is an upper-case letter in range $['A', 'O']$.

$\begin{matrix} x & x & x & x \\ 3 & 2 & 1 & 0 \end{matrix} = 0111$, since the last 4 bits in both p_2 and p_3 are not equal then we can't determine exactly which letter or digit correspond to the plaintext.

$p_1 \rightarrow$ lower-case letter $['p', 'z']$.

$p_2 \rightarrow$ digit.

$p_3 \rightarrow$ upper-case letter $['A', 'O']$.

Question #7:

Question 7

$$\text{maj}(x_8, y_{10}, z_{10}) = (1, 1, 0) = 1$$

$x_8 = 1$ then x steps 8

$$t = 0 \oplus 1 \oplus 1 \oplus 0 = 0$$

$$x_i = x_{i-1}, \quad x_0 = t = 0$$

$$x = 010101010101011$$

$y_{10} = 1$ then y steps.

$$t = 1 \oplus 1 = 0$$

$$y_i = y_{i-1}, \quad y_0 = 0$$

$$y = 0110011000110110001001$$

$z_{10} \neq 1$ then z does not step.

$$z = 11100101110000011000011$$

$$1^{\text{st}} \text{ keystream bit} = x_{18} \oplus y_{21} \oplus z_{22} = 1 \oplus 1 \oplus 1 = 1$$

$$\text{maj}(x_8, y_{10}, z_{10}) = (0, 1, 0) = 0$$

$x_8 = 0$ then x steps.

$$t = 1$$

$$x_i = x_{i-1}, \quad x_0 = 1$$

$$x = 10101010101010101$$

$y_{10} \neq 0$ then y does not step

$$y = 0110011000110110001001$$

$z_{10} = 0$ then z steps.

$$t = 1$$

$$z_i = z_{i-1}, \quad z_0 = 1$$

$$z = 111100101100000110001$$

$$2^{\text{nd}} \text{ keystream bit} = x_{18} \oplus y_{21} \oplus z_{22} = 1 \oplus 1 \oplus 1 = 1$$

$$\text{maj}(1, 1, 1) = 1$$

$x_8 = 1$ then x steps

$$t = 0$$

$$x = 010101010101010101010$$

$y_{10} = 1$ then y steps.

$$t = 1$$

$$y = 1011001100011011000100$$

$z_{10} = 1$ then z steps.

$$t = 1$$

$$z = 11111001011100000110000$$

$$3^{\text{rd}} \text{ keystream bit} = 0 \oplus 0 \oplus 0 = 0$$

$$\text{maj}(0, 0, 1) = 0$$

$x_8 = 0$ then x steps.

$$t = 1$$

$$x = 10101010101010101$$

$y_{10} = 0$ then y steps

$$t = 0$$

$$y = 01011001100011011000100$$

$z_{10} \neq 0$ then z does not step.

$$z = 11111001011100000110000$$

$$4^{\text{th}} \text{ keystream bit} = 1 \oplus 0 \oplus 0 = 1$$

$\Rightarrow$ the 4 keystream bits are 1101

Question #8:

Question 8

a) in order to find round key for round 1 and round key for round 2 we need to find the output for each table F_{k_1} and F_{k_2} in a way that have fewer operations than exhaustive search means that total number of operation should be $< 2^{56}$ iteration.

F_{k_1}

Assume that the left size is fixed $L_0 = 0$ and the right side $R_0 = 0, 1, \dots, 2^{32-1}$.

$$R_1 = L_0 \oplus F(k_1, R_0)$$

$$\text{but } L_2 = R_1$$

$$L_2 = L_0 \oplus F(k_1, R_0) \Rightarrow F(k_1, R_0) = L_0 \oplus L_2$$

$$L_0 = 0 \Rightarrow F(k_1, R_0) = L_2$$

then searching in the table for R_0 should take 2^{32} steps to find $F(R_0) = L_2$

for the second table

$$R_2 = L_1 \oplus F(k_2, R_1)$$

$$\text{but } L_1 = R_0$$

$$R_2 = R_0 \oplus F(k_2, R_1)$$

$$F(k_2, R_1) = R_2 \oplus R_0$$

if we suppose that $R_0 = 0$ and $L_0 = 0, 1, \dots, 2^{32-1}$.

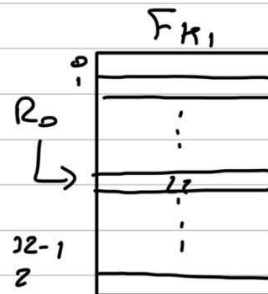
$$F(k_2, R_1) = R_2 \quad \text{but } R_1 = L_0 \oplus F(k_1, (0)).$$

$$F(k_2, L_0 \oplus \underbrace{F(k_1, (0))}_{\text{this is constant}}) = R_0$$

this is
constant

then to find the output of the table f_{k_2} we needs 2^{32} steps.

$$\text{total operation} = \frac{32}{2} + \frac{32}{2} = \frac{33}{2} < \frac{56}{2} \quad \text{good!!}$$



b) we found that :

$$R_1 = L_0 \oplus F(k_1, R_0) = L_2$$

$$R_2 = R_0 \oplus F(k_2, L_0 \oplus F_{k_1}(R_0))$$

$$C = L_2, R_2 = \left((L_0 \oplus F_{k_1}(R_0)), (R_0 \oplus F_{k_2}(R_1)) \right)$$

lets take some plain/cipher text pairs to see if we can distinguish DES2 from a random permutation.

$$P_0 = (A_0, B_0) \quad , \quad P_1 = (A_1, B_0) \quad \begin{matrix} A_0 \neq A_1 \\ B_0 = B_0 \end{matrix}$$

$$C_0 = (X_0, Y_0) \quad , \quad C_1 = (X_1, Y_1)$$

lets take the xor of the two halves of cipher :

$$X_0 \oplus X_1 = (A_0 \oplus F_{k_1}(B_0)) \oplus (A_1 \oplus F_{k_1}(B_0))$$

$$= A_0 \oplus A_1 \oplus \cancel{F_{k_1}(B_0)} \oplus \cancel{F_{k_1}(B_0)}$$

$$X_0 \oplus X_1 = A_0 \oplus A_1$$

Since the xor of two cipher texts gave me the corresponding xor of the plain texts, then the algorithm used is PFS 2 not a random permutation.

Question #9:

Question 9

$$a) s_{i,0} = (B4, 52, E0, AE)_{16}$$

$$\text{InvMix Matrix} \begin{pmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{pmatrix} * \begin{pmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \end{pmatrix}$$

$$\begin{pmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \end{pmatrix} = \begin{pmatrix} B4 \\ 52 \\ E0 \\ AE \end{pmatrix}$$

Third element is :

$$= 0D \times B4 \oplus 09 \times 52 \oplus 0E \times E0 \oplus 0B \times AE$$

$$= 85 \oplus F4 \oplus 37 \oplus EE$$

$$\begin{aligned} 00 \times B4 &= (x^3 + x^2 + 1) \times (x^7 + x^5 + x^4 + x^2) \\ &= \cancel{x^7} + \cancel{x^5} + \cancel{x^4} + x^2 + \cancel{x^9} + \cancel{x^7} + \cancel{x^6} + \cancel{x^4} + x^{10} + \cancel{x^8} \\ &\quad + \cancel{x^7} + \cancel{x^5} \\ &= x^{10} + x^9 + x^8 + x^7 + x^6 + x^2 \end{aligned}$$

$$= \cancel{x^7} + \cancel{x^5} + \cancel{x^4} + x^2 + \cancel{x^9} + \cancel{x^7} + \cancel{x^6} + \cancel{x^4} + x^{10} + x^8$$

$$+ x^7 + \cancel{x^5}$$

$$= x^{10} + x^9 + x^8 + x^7 + x^6 + x^2$$

reduction %

$$\begin{array}{r}
 x^2 + x + 1 \\
 \hline
 \begin{array}{l}
 \text{8} \quad \text{4} \quad \text{3} \\
 x + x + x + x + 1
 \end{array}
 \left\{
 \begin{array}{l}
 \cancel{x^{10}} + \cancel{x^9} + \cancel{x^8} + \cancel{x^7} + \cancel{x^6} + \cancel{x^5} + \cancel{x^4} + \cancel{x^3} + \cancel{x^2} \\
 \phantom{\cancel{x^{10}}} + \cancel{x^6} + x^5 + x^3 + \cancel{x^2}
 \end{array}
 \right. \\
 \hline
 \begin{array}{l}
 \text{9} \quad \text{8} \quad \text{7} \quad \text{5} \quad \text{3} \\
 \cancel{x} + \cancel{x} + \cancel{x} + \cancel{x} + x^3
 \end{array} \\
 \begin{array}{l}
 \text{9} \quad \text{5} \quad \text{4} \quad \text{2} \\
 \cancel{x} + \cancel{x} + x^4 + x^2 + x
 \end{array} \\
 \hline
 \begin{array}{l}
 \text{8} \quad \text{7} \quad \text{4} \quad \text{3} \quad \text{2} \\
 \cancel{x} + \cancel{x} + \cancel{x} + \cancel{x} + x^2 + \cancel{x}
 \end{array} \\
 \begin{array}{l}
 \text{8} \quad \text{4} \quad \text{3} \\
 \cancel{x} + \cancel{x} + \cancel{x} + \cancel{x} + 1
 \end{array} \\
 \hline
 \begin{array}{l}
 \text{7} \quad \text{2} \\
 x^7 + x^2 + 1
 \end{array}
 \end{array}$$

$$x^7 + x^2 + 1 \Rightarrow 85$$

$$\begin{aligned}
 09 \times 52 &= (x^3 + 1) \times (x^6 + x^4 + x^2) \\
 &= x^6 + \cancel{x^4} + x^2 + x^9 + x^7 + \cancel{x^5} \\
 &= x^9 + x^7 + x^6 + x^2
 \end{aligned}$$

reduction 8

$$\begin{array}{r}
 \begin{array}{c} x^8 \quad x^4 \quad x^3 \\ x^8 + x^4 + x^3 + x + 1 \end{array} \overline{) \begin{array}{c} x^9 + x^7 + x^6 + x^5 \\ \underline{x^9 + x^5 + x^4 + x^2 + x} \\ x^7 + x^6 + x^5 + x^4 + x^2 \end{array}}
 \end{array}$$

$$x^7 + x^6 + x^5 + x^4 + x^2 \Rightarrow F4$$

$$0Ex E_0 = (x^3 + x^2 + x) \times (x^7 + x^6 + x^5)$$

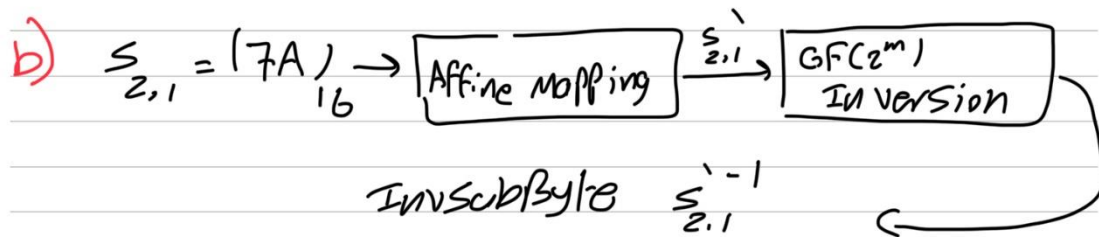
$$= x^8 + \cancel{x^7} + x^6 + \cancel{x^9} + \cancel{x^8} + \cancel{x^7} + x^{10} + \cancel{x^9} + \cancel{x^8}$$

$$= x^{10} + x^8 + x^6$$

reduction 8

$$\begin{array}{r}
 \begin{array}{c} 8 \quad 4 \quad 3 \\ x + x + x + x + 1 \end{array} \overline{) \begin{array}{c} 2 \\ x + 1 \end{array}} \\
 \underline{\begin{array}{c} 10 \quad 8 \quad 6 \\ \cancel{x} + x + \cancel{x} \end{array}} \\
 \begin{array}{c} \cancel{x^{10}} + \cancel{x^6} + x^5 + x^3 + x^2 \\ \hline \cancel{x^8} + x^5 + \cancel{x^3} + x^2 \\ \cancel{x^8} + x^4 + \cancel{x^3} + x + 1 \\ \hline x^5 + x^4 + x^2 + x + 1 \end{array}
 \end{array}$$

$$x^5 + x^4 + x^2 + x + 1 \Rightarrow 37$$



$$(7A)_{16} \rightarrow 0111 \ 1010$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \pmod{2}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = 1011 \ 1100$$

(BC)
16

now the inverse of $(BC)_{16}$

$$(BC)_{16} = x^7 + x^5 + x^4 + x^3 + x^2$$

i	q_{i-1}	r_i	s_i	t_i
1		$x^8 + x^4 + x^3 + x + 1$	1	0
2		$x^7 + x^5 + x^4 + x^3 + x^2$	0	1
3	x	$x^6 + x^5 + x + 1$	1	x
4	$x + 1$	$x^4 + x^3 + 1$	$x + 1$	$x^2 + x + 1$
5	x^2	$x^2 + x + 1$	$x^3 + x + 1$	$x^4 + x^3 + x^2 + x$
6	$x^2 + 1$	x	$x^5 + x^2$	$x^6 + x^5 + 1$
7	$x + 1$	1	$x^6 + x^5 + x^2 + x + 1$	$x^7 + x^5 + x^4 + x^3 + x^2 + x + 1$
8		0		

$$x^7 + x^5 + x^4 + x^3 + x^2 + 1 = 10111101$$

$$\text{InvSubByte} = (BD)_{16}$$

c) K6 :

98 0F 71 AF 15 C9 47 D9 0C B7 E8 59 D6 7F 67 AD

g Function on :

D6 7F 67 AD

7F 67 AD D6

D2 ⊕ 40

= 92

⊕ D2 85 95 F6
RC[7]

92 85 95 F6

Round 7

$w_{71} = 98 \ 0F \ 71 \ AF \ 92 \ 85 \ 95 \ F6 \oplus = 0A \ 8A \ E4 \ 59$

$w_{72} = 15 \ C9 \ 47 \ D9 \ 0A \ 8A \ E4 \ 59 \oplus = 1F \ 43 \ A3 \ 80$

$w_{73} = 0C \ B7 \ E8 \ 59 \ 1F \ 43 \ A3 \ 80 \oplus = 13 \ F4 \ 4B \ D9$

$w_{74} = D6 \ 7F \ 67 \ AD \ B \ F4 \ 4B \ D9 \oplus = C5 \ 8B \ 2C \ 74$

g Function on 8

DA C8 8F F4

E8 ⊕ 10

C8 8F F4 DA

= C8

E8 73 BF 57

⊕ RC[5]

round 5

C8 73 BF 57

$$\omega_0 = \omega_4 \oplus g(\omega_3) = \begin{array}{cccc} 98 & 0F & 71 & AF \\ C8 & 73 & BF & 57 \end{array} \oplus$$

$$= 50 \ 7C \ CE \ F8$$

$$\omega_1 = \omega_5 + \omega_4 = \begin{array}{cccc} 98 & 0F & 71 & AF \\ 15 & C9 & 47 & D9 \end{array} \oplus$$

$$= 8D \ C6 \ 36 \ 76$$

$$\omega_2 = \omega_6 \oplus \omega_5 = \begin{array}{cccc} 15 & C9 & 47 & D9 \\ 0C & B7 & E8 & 59 \end{array} \oplus$$

$$= 19 \ 7E \ AF \ 80$$

$$\omega_3 = \omega_7 \oplus \omega_6 = \begin{array}{cccc} 06 & 7F & 67 & AD \\ 0C & B7 & E8 & 59 \end{array} \oplus$$

$$= DA \ C8 \ 8F \ F4$$